

Project Name

Cardiovascular Disease Prevention Platform

Project Overview

Project Type: Diploma

Domain: Web Development, Healthcare Informatics, Data Analytics

Student Information

- **Full Name:** Darya Korbut
- **Group:** JS/TS
- **Supervisor:** Alevtina Gourinovich

Project Description

This project focuses on the development of a web-based knowledge system for cardiovascular disease prevention. The system combines a React frontend, an Express/Prisma backend, and a PostgreSQL database to provide standardized, bilingual (EN/RU) disease cards with consistent descriptions, risk factors, symptoms, and prevention guidance.

The project emphasizes production-ready architecture with containerization, comprehensive testing, observability, and API-first design suitable for future expansion.

The research component includes analysis of real-world cardiovascular datasets (UCI Heart Disease, Statlog, Cleveland, Hungary) containing clinical data from actual patients. This data-driven approach ensures that the prevention recommendations and risk factor information are grounded in validated medical research rather than theoretical assumptions.

The project carries significant social value by making evidence-based cardiovascular health information accessible to the general public, potentially contributing to early disease awareness and prevention in communities where such knowledge may be limited.

Links

- **Source Code Repository:** <https://github.com/Doshirac/cvd-platform>
- **API Documentation:** <https://app.gitbook.com/invite/YJjvuHTqbLlvmjZEzQci/J36lOoZ1hCDc4L4xnq64>
- **Database Documentation:** <https://docs.google.com/document/d/1FeM152YZaFifvUPv-9Nsqt8NmtA2ZMBfu1xeoPYsN0/edit?usp=sharing>
- **UI/UX Documentation:** https://docs.google.com/document/d/19rmvxAH_jqToaFLBTyVxYtAc306CxIKx-DLcIrTWUuQ/edit?usp=sharing
- **Containerization Documentation:** https://docs.google.com/document/d/1yJLejGVrdJQ4Gq0f72P750_r9pzv-XMS-WSrUH_jQEI/edit?usp=sharing
- **BA Requirements Documentation:** <https://docs.google.com/document/d/16HgcMeCpzlgXRnMC0qnYaDRJgjW5>

Selected Evaluation Criteria

- Frontend Architecture & UX Delivery
- Backend Architecture & API Delivery
- Database Design & Data Integrity
- Observability (Logging + Error Monitoring)
- Research Pipeline & Analytics Artifacts
- Deployment & Local Environment
- API Documentation
- Data Analysis of Real Datasets

Project Status

- Production-ready MVP prototype
- Backend API fully functional with bilingual support
- No continuous data updates (read-only knowledge base)

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Features & Requirements

Core Features (Epics)

Epic ID	Feature	Description	Priority
E1	Browse Disease Library	Discover diseases via list view	Must
E2	Search Diseases	Find diseases by name/keywords	Must
E3	Filter by Category	Narrow results by category/risk/severity	Must
E4	View Disease Card	Read standardized disease card	Must
E5	View Research Cards	Read-only research summaries	Should
E6	Switch Language EN/RU	Toggle English/Russian	Must
E7	View References	View sources/datasets with provenance	Should

Functional Requirements - User Stories

User Story 1 - Browse Disease Library

As a User

I want to see a list of cardiovascular diseases

So that I can quickly find conditions to read.

Acceptance Criteria: - List loads $\leq 2s$ with ≥ 25 diseases - Each item links to its card - Clear empty/error states

User Story 2 - Search Diseases

As a User

I want to search by disease name or keywords

So that I can find a specific disease.

Acceptance Criteria: - Results appear $< 1s$ - Matches are highlighted - No-results state shown

User Story 3 - Filter by Category

As a User

I want to apply filters

So that I can narrow the list to what I need.

Acceptance Criteria: - Filters update results instantly - “Clear all” resets state - URL reflects active filters (shareable)

User Story 4 - View Disease Card

As a User

I want to open a disease card

So that I can read description, symptoms, and prevention.

Acceptance Criteria: - Sections visible: Description, Primary/Secondary symptoms, Prevention - Loads $\leq 2s$ - Sources/references visible

User Story 5 - Switch Language

As a User

I want to switch between English and Russian

So that I can read in my preferred language.

Acceptance Criteria: - UI labels and content switch - Choice persists across sessions

Use Case Diagram

[View Diagram](#)

Non-Functional Requirements

- **Performance:** LCP <2.5s; API p95 <300ms; 25-50 concurrent users
- **Security:** HTTPS, rate limiting, input validation, no PII/PHI
- **Accessibility:** Keyboard navigation, ARIA labels, medical disclaimer
- **Scalability:** Stateless API, cached assets
- **Usability:** Responsive UI, Dark/Light theme, EN/RU localization
- **Reliability:** Daily backups, <=60min recovery target
- **Compatibility:** Chrome/Firefox/Safari (latest 2 versions), iOS 16+/Android 11+
- **Deployment:** Docker Compose, OpenAPI documentation

Compliance

- **GDPR:** No PII/PHI; Privacy Notice required
- **Medical Disclaimer:** Informational only, not medical advice
- **Data Licensing:** Respect open-data licenses (WHO/EU/national)
- **Terms of Use:** No warranty, liability limitations

Project Overview

Business context, goals, and requirements for the CVD knowledge platform.

Contents

- [Original Documentation](#)
- [Problem Statement & Goals](#)
- [Stakeholders & Users](#)
- [Scope](#)
- [Features](#)

Executive Summary

Web platform consolidating fragmented CVD information into a standardized, referenced disease library for clinicians, educators, and public-health analysts.

Architecture: React SPA + Express/Prisma + PostgreSQL + Docker

Key Highlights

Aspect	Description
Problem	Inconsistent structure and weak source transparency in CVD info

Aspect	Description
Solution	Standardized disease cards with references + dataset-driven analytics
Target Users	Clinicians, educators, public-health analysts
Key Capabilities	Search, filters, EN/RU localization, references page
MVP Outcomes	25 disease cards; 2 datasets with provenance

Problem Statement & Goals

Context

Read-only CVD knowledge platform for education and public-health exploration. Focuses on evidence-linked summaries and dataset-driven indicators (not diagnosis or clinical decision support).

Vision Statement

Build a clear, reliable platform with standardized cards for cardiovascular diseases (description, symptoms, prevention) and basic statistics such as prevalence and key indicators from trusted data sources. The platform helps clinicians, educators, and public-health analysts distinguish primary symptoms from secondary factors, and it is designed to be extendable beyond the diploma scope.

Problem Statement

Who:

Clinicians, medical educators, public-health analysts, and readers needing structured CVD information.

What:

CVD information is fragmented, inconsistently structured, and lacks clear separation of primary symptoms vs. secondary factors.

Why:

Users spend time reconciling disparate sources, leading to slower information retrieval and unclear prevention guidance.

Business Goals

Goal	Description	KPI
Content Coverage	Standardized disease cards with references	25 cards
Data Transparency	Dataset provenance (source + last updated)	2 datasets
Discoverability	Fast search and filters, bilingual UI	EN/RU support
Maintainability	Reproducible system	Docker, OpenAPI
Privacy	No personal data processing	No PII/PHI

Objectives (MVP Targets)

- Content: 25 disease cards
- Data: 2 official datasets with provenance
- UX: Search + filters + EN/RU toggle
- API: OpenAPI documentation

Non-Goals

The project does **not** aim to: - Provide diagnosis/treatment advice or symptom checker - Implement user accounts - Build real-time BI stack or ML prediction - Develop native apps/offline mode

Pain Points

#	Pain Point	Severity	Current Workaround
1	Information is scattered across many sources and not standardized (terminology, structure, completeness).	High	Manual search in guidelines/articles; personal notes; ad-hoc comparisons.
2	Primary symptoms vs. secondary factors (risk/prevention-related) are often mixed, making interpretation and teaching harder.	High	Rewriting content into personal templates; discussing with colleagues; cross-checking multiple sources.
3	Lack of a single place to read curated research summaries and supporting context linked to transparent sources.	Medium	Searching publications manually; bookmarking links; assembling summaries in notes.
4	Unclear data provenance (source, last updated, licensing) reduces trust and reusability.	Medium	Checking dataset websites and PDFs manually; storing links outside the workflow.
5	Hard to quickly find relevant diseases and compare them without strong search/filters.	Medium	Browser search; bookmark lists; manual scanning.

Project Scope

In Scope

The system includes functionality for: - Disease library (≥ 25 entries) with standardized content and references - Search and filters for discovery - EN/RU locale support - Sources/references page with provenance (source + last updated) - Research summaries/cards (read-only) - API documentation and containerized deployment

Out of Scope

Excluded functionality: - Medical diagnosis/treatment advice or symptom checker - User accounts (bookmarks, comments, forums) - Advanced BI stack or real-time dashboards - ML risk prediction or personalized recommendations - Native mobile apps and offline mode

Assumptions

- Public datasets available for academic use with citation
- No PII/PHI collected or processed
- Docker-friendly hosting environment available
- Clinical consultant available for content review

Constraints

Category	Description
Time	~4 months (diploma schedule)
Resources	Single developer (FE/BE/data/content)
Technology	React, Node.js (Express), Python, PostgreSQL, Docker
Budget	Open-source tools and open data only
Compliance	Dataset licenses, GDPR-aware setup

Stakeholders & Users

Target Audience

User Group	Description	Primary Needs
Clinicians / Medical students	Structured CVD information users	Fast lookup, consistent terminology, references
Educators / Researchers	Teaching materials users	Reliable citations, exportable summaries
Analysts / Public health readers	Data exploration users	Analytics visuals, source transparency

Stakeholder Analysis Matrix

Stakeholder	Role	Interest	Influence	Engagement
Product Owner & Engineer	Implementer	High	High	Weekly milestones
Academic Supervisor	Evaluator	Medium	High	Control checkpoints
Clinical Consultant	Content reviewer	High	High	2-3 review sessions
End Users	Platform users	High	Medium	Usability tests
Data Source Owners	Data providers	Medium	Medium	License compliance

Stakeholder Expectations

- Accuracy and reliable references
- Consistent terminology and structure
- Clear symptom/risk-factor separation
- Source transparency (provenance + last updated)
- Academic methodological rigor

Technical Implementation

Technical architecture, design decisions, and implementation details of the CVD Platform.

Contents

- [Tech Stack](#)
- [Criteria Documentation](#)
- [Deployment](#)

Original Documentation

- [Backend Documentation](#)
- [Database Documentation](#)
- [Containerization Documentation](#)
- [API Documentation](#)

Architecture

Diagrams

- [High-Level Architecture](#)
- [ER Diagram](#)
- [Deployment Diagram](#)

System Components

Component	Technology	Description
Frontend	React + Vite	SPA with theming, routing, error boundaries
Backend	Express + Prisma	REST API with pagination, filtering, i18n
Database	PostgreSQL 16	Normalized content store
Cache	Redis 7	Read caching layer
Monitoring	Sentry	Error tracking (optional)

Data Flow

[User] → [React SPA] → [Express API] → [Redis Cache] → [PostgreSQL]
(cache miss) [Prisma] → [DB]

Key Decisions

Decision	Rationale	Alternative
React + Vite SPA	Fast iteration, simple deployment	Next.js SSR
Express + Prisma	Lightweight, type-safe API	Fastify, NestJS
Normalized schema	Data integrity, M:N, i18n	Single-table MVP
Redis caching	Reduces DB load	In-memory cache
Sentry + logs	Practical debugging	Console-only

Criteria Documentation

The evaluation criteria ADRs are located in [criteria/](#) and include:

- [Frontend](#)
- [Backend](#)
- [Database](#)

- [Research](#)
- [Deployment](#)
- [Observability](#)
- [Testing](#)

Technology Stack

Stack Overview

Layer	Technology	Version	Justification
Frontend	React + TypeScript + Vite	React 19.1, TS 5.8, Vite 6.3	Fast SPA dev/build
UI Styling	SCSS Modules + Sass	Sass 1.89	Lightweight theming
Backend	Node.js + Express	Node 20, Express 5.1	REST API performance
Database	PostgreSQL	16	Relational integrity
ORM	Prisma	6.19	Type-safe data access
Cache	Redis + ioredis	Redis 7	Read caching
Analytics	Python ETL	3.x	Deterministic outputs
Deployment	Docker Compose	-	Multi-service orchestration

Key Decisions

Decision	Rationale	Alternative Rejected
React + Vite	Fast iteration, SPA routing, theming	Next.js (SSR complexity)
Express + Prisma	Lightweight API, type-safe DB	Django (heavier stack)
Redux Toolkit	Centralized data lifecycle	Local state only
Normalized schema	Data integrity, M:N relations, i18n	Single denormalized table
Python ETL	Cacheable visuals, no BI dependencies	Power BI/live charts

Development Tools

Tool	Purpose
VS Code	IDE with ESLint/Prettier
Git + Husky	Version control, pre-commit hooks
Jest + Playwright	Unit/E2E testing
Prisma CLI	Migrations, seeding
OpenAPI + Storybook	API/component documentation
Docker Compose	Local environment

External Services

Service	Purpose
Sentry	Error monitoring
WHO/Eurostat/CDC	Open health datasets

Backend Architecture & API Delivery

Original documentation: <https://docs.google.com/document/d/1UAZhPh1Rj8UjTQB2ZCpyM1jcoRafcPedou2hYEtP60/edit>

Decision

Status: Accepted | **Date:** 2026-01-05

TypeScript backend on Node.js + Express with layered architecture (controllers -> services -> repositories), Prisma ORM, Redis caching, centralized error handling, and OpenAPI docs.

Alternative	Why Not Chosen
Fastify	Express fits existing tooling
NestJS	Too much overhead for MVP
FastAPI	Would require two-stack split

Project Structure

```

backend/src/
  app/          # Express composition
  routes/       # /health, /diseases, /sources
  disease/      # Domain module (controller/service/repo/cache)
  source/       # Domain module
  config/       # Env, swagger, Sentry
  middlewares/  # notFound + error
  errors/       # ApiError factory
  cache/        # Redis client
  utils/        # Logger + request logger

```

Requirements

#	Requirement	Status
1	Express middleware + routes	Done
2	REST endpoints (diseases/sources/health)	Done
3	Pagination/filtering/search	Done
4	Locale-aware responses (en/ru)	Done
5	OpenAPI documentation	Done
6	Centralized error middleware	Done
7	Redis cache	Done
8	Consistent DI	Done

Limitations

Limitation	Solution
DI inconsistently applied	Move to container bindings
Inconsistent empty response shapes	Return consistent envelope

Conclusion

The CVD Platform Backend delivers a production-ready, modular API for cardiovascular disease data with a focus on reliability, observability, and developer experience.

Key Achievements: - Modular monolith architecture with clear domain boundaries (disease, source, cache, monitoring) and dependency injection - Full containerization via Docker with multi-stage builds, health checks, resource limits, and graceful shutdown - Comprehensive observability: structured JSON logging, Sentry error tracking, health endpoints, and centralized error handling - High test coverage (87%+ lines) with Jest, enforced code quality via ESLint/Prettier/Husky pre-commit hooks - Bilingual support through Prisma translation tables, Redis caching for performance, and OpenAPI/Swagger documentation

This implementation provides query-optimized, language-aware access to cardiovascular disease data and is ready for integration with frontend applications, analytics dashboards, and future microservices expansion.

Database Design & Data Integrity

Original documentation: <https://docs.google.com/document/d/1FeM152YZaFifvUPv-9Nsqt8NmtA2ZMBfu1xeoPYsN0/edit>

Decision

Status: Accepted | **Date:** 2026-01-05

PostgreSQL with Prisma ORM using normalized schema: core entities (Disease, Symptom, RiskFactor, Source), M:N junction tables with metadata, and translation tables keyed by locale.

Alternative	Why Not Chosen
Single denormalized table	Poor integrity, hard to maintain
MongoDB	Data is relational
Raw SQL	Prisma offers type safety

Structure

```
backend/prisma/
  schema.prisma  # Canonical schema
  migrations/    # Migration history
  seed.ts       # Seed script
```

Requirements

#	Requirement	Status
1	PostgreSQL for structured content	Done
2	Type-safe ORM with migrations	Done
3	Bilingual support (EN/RU)	Done
4	M:N disease-symptom with metadata	Done
5	M:N disease-risk factor with direction	Done
6	Integrity constraints + indexes	Done

#	Requirement	Status
7	Sources for attribution	Done
8	Seed process	Done

Limitations

Limitation	Solution
Source not linked to diseases	Add DiseaseSource relation
Full-text search may be slow	Add tsvector + GIN indexes

Conclusion

The Cardiovascular Disease Knowledge Base database provides a robust, normalized foundation for clinical and public-health data on cardiovascular conditions.

Key Achievements: - Comprehensive entity structure (diseases, symptoms, risk factors) - Strict referential integrity with clear indexes and constraints - Bilingual (localized) design via translation tables and SQL views - Automated versioning and migrations through Prisma and Git - Extensible for future data layers (raw -> staging -> mart)

This implementation supports consistent, query-optimized, and language-aware access to cardiovascular disease data. It can be integrated into an analytical dashboard.

Deployment & DevOps

Original documentation: https://docs.google.com/document/d/1yJLejGVrdJQ4Gq0f72P750_r9pzv-XMS-WSrUH_jQEI/edit?usp=sharing

Overview

The platform runs as three services via Docker Compose: API (Node/Express), PostgreSQL, and Redis. This provides a reproducible local environment and a production-like deployment model.

Diagram

[Deployment diagram](#)

How to Run (Local)

1. Create a `.env` file (or set environment variables) for the backend.
2. Run `docker compose up --build`.
3. Verify health: `GET /api/health` returns 204.

Environment Variables

Variable	Default	Required	Description	Example
PORT	4000	Yes	The port on which the Express API server listens. Used by the API container and referenced in health checks.	4000
NODE_ENV	development	No	Runtime environment mode. Set to production for production builds. Affects logging verbosity, error detail exposure, and certain middleware behavior.	production
DATABASE_URL	-	Yes	Full PostgreSQL connection string used by Prisma. Format: <code>postgresql://USER:PASSWORD@HOST:PORT/DATABASE</code>	postgresql://cvd_user:password@localhost:5432/cvd_db
POSTGRES_USER	-	Yes	PostgreSQL username. Used by the db container in docker-compose.yml.	postgres
POSTGRES_PASSWORD	-	Yes	PostgreSQL password. Used by the db container.	password
POSTGRES_DB	-	Yes	PostgreSQL database name. Used by the db container.	cvd_db
REDIS_HOST	localhost	Yes	Hostname or IP of the Redis server. Set to redis when running via Docker Compose.	redis
REDIS_PORT	6379	Yes	Port on which Redis is listening.	6379

Variable	Default	Required	Description	Example
SENTRY_DSN	-	No	Sentry project DSN for error tracking and performance monitoring. If not provided, Sentry integration is disabled.	<code>https://...</code>
CORS_ORIGIN	*	No	Allowed origin(s) for CORS requests. Set to your frontend URL in production (e.g., <code>https://your-frontend.com</code>).	<code>http://localhost:5173</code>
API_URL	-	No	Base URL of the backend REST API, used by the frontend and tooling to build requests (e.g. <code>http://localhost:4000/api</code>). Typically matches the external address of the <code>/api</code> prefix in development and production.	<code>http://localhost:4000/api</code>
CLIENT_URL	-	No	Base URL of the frontend application that consumes this API (e.g. <code>http://localhost:5173</code>). Can be used for redirects, links in error messages, or security checks when needed.	<code>http://localhost:5173</code>

Variable	Default	Required	Description	Example
LOG_LEVEL	info	No	Minimal log level for the application logger (e.g. info , debug , warn , error). This controls verbosity of JSON logs and is useful to reduce noise in production while keeping more detailed output in development.	debug

Notes

- Health checks are configured for API/DB/Redis in `docker-compose.yml`.
- Persistent data is stored in Docker volumes for PostgreSQL and Redis.

Conclusion

The CVD platform containerization implementation demonstrates a production-ready Docker Compose architecture with three core services: backend API (cvd-api), PostgreSQL database (db-1), and Redis cache (redis-1). The configuration adheres to containerization best practices including multi-stage builds, resource limits, health checks, and persistent volume management.

Key Achievements: - Multi-stage Dockerfile reduces API image size from ~1.5 GB to 891 MB by eliminating build tools and devDependencies from the final image - Named volumes (pgdata, redisdata) ensure data persistence across container restarts and deployments - Health checks monitor service availability every 30 seconds with automatic recovery after 3 consecutive failures - Resource limits (CPU: 0.5 cores API/DB, 0.25 cores Redis; Memory: 512 MB API/DB, 256 MB Redis) prevent resource exhaustion on shared infrastructure - Non-root user execution enhances security posture

Frontend Architecture & UX Delivery

Decision

Status: Accepted | **Date:** 2026-01-05

React + TypeScript SPA using Vite. React Router for navigation, Redux Toolkit for shared data flow, Error Boundary + Sentry for crash handling, SCSS modules for theming.

Alternative	Why Not Chosen
Next.js (SSR/ISR)	Extra complexity for read-only library
No global store	Harder cross-page caching
UI framework (MUI)	Already using SCSS modules

Project Structure

```
frontend/src/
  app/
    App.tsx          # BrowserRouter + providers
    layouts/MainLayout/ # Header + Outlet shell
    providers/        # ErrorBoundary, Router, StoreProvider
  pages/              # Route pages (Error/NotFound implemented)
  shared/
    api/              # API client + slices/thunks
    context/          # Theme context
    hooks/            # useTheme and helpers
    ui/               # Shared UI components
  assets/
```

Requirements

#	Requirement	Status
1	SPA entrypoint and routing	Partially
2	Centralized layout shell	Done
3	Shared store for API data	Done
4	Error handling	Partially
5	Theme context	Done
6	Main library pages wired	Not started
7	Research cards route	Not started

Limitations

Limitation	Solution
Core pages not mounted	Implement MainPage, SourcePage, DiseasePage
SPA SEO limited	Add sitemap/meta tags

Conclusion

The CVD Platform Frontend delivers a modern, maintainable React SPA for browsing cardiovascular disease content with strong developer experience and UX foundations.

Key Achievements: - React 19 + TypeScript + Vite stack with fast HMR and optimized builds - Redux Toolkit for centralized state management with RTK Query scaffold for API data - 13+ reusable UI components (Header, Button, Input, Loader, ThemeToggle, LanguageSwitcher, etc.) - Dark/Light theming via SCSS modules and ThemeContext - Error Boundary + Sentry integration for crash handling and monitoring - Playwright E2E tests (error-page, not-found-page, theme-toggle) and Jest unit test setup - Storybook for component documentation and visual testing - Code quality enforced via ESLint, Prettier, Stylelint, and Husky pre-commit hooks

This implementation provides a scalable foundation for disease library UI with theming, i18n readiness, and comprehensive error handling.

Observability (Logging + Error Monitoring)

Decision

Status: Accepted | Date: 2026-01-05

Sentry for error monitoring (frontend + backend). Winston for backend logs, Morgan for HTTP request logs. Frontend Error Boundary for crash handling.

Alternative	Why Not Chosen
Full APM suite	Cost + setup overhead
Console-only logs	No aggregation/structure
No client monitoring	Silent UI failures

Structure

```
frontend/src/main.tsx          # Sentry init
frontend/src/app/providers/ErrorBoundary/
backend/src/config/instrument.js # Sentry init
backend/src/utils/logger.ts     # Winston
backend/src/utils/requestLogger.ts # Morgan
backend/src/middlewares/errorMiddleware.ts
```

Requirements

#	Requirement	Status
1	Frontend crash fallback	Partially
2	Frontend error reporting	Partially
3	Backend error reporting	Done
4	Backend structured logs	Done

Limitations

Limitation	Solution
Sentry may capture extra context	Disable default PII collection
Monitoring disabled if DSN missing	Require DSN for staging/prod

Conclusion

The observability implementation provides comprehensive monitoring and debugging capabilities for both frontend and backend systems.

Structured Logging: - Winston logger with configurable log levels (LOG_LEVEL env variable) - JSON-formatted logs to stdout/stderr for Docker compatibility - Log entries include: level, message, timestamp, service identifier, context fields - Morgan HTTP request logging (method, path, status, response time) - Docker json-file driver integration for centralized logging (Loki, ELK compatible)

Error Tracking: - Sentry (@sentry/node, @sentry/react) for real-time exception monitoring - Automatic capture of unhandled exceptions and promise rejections - Manual Sentry.captureException() calls in catch blocks for context-rich reports - Performance monitoring for transaction tracking and bottleneck identification - Optional via SENTRY_DSN environment variable - Debug endpoint (/api/debug-sentry) for integration testing

Global Error Handling: - Centralized Express error middleware catches all controller/service errors - Automatic logging with full context (message, stack, path, method) - Automatic Sentry reporting without manual catch blocks - Structured JSON responses (success: false, message, errors[], stack in dev) - Custom ApiError class for semantic HTTP status codes - 404 Not Found middleware for undefined routes

This implementation enables faster debugging, clearer demo reliability, and comprehensive error visibility across the full stack.

Research Pipeline & Analytics Artifacts

Decision

Status: Accepted | **Date:** 2026-01-05

Python ETL under analysis/ with datasets committed for reproducibility. Generate static artifacts (SVG/PNG) for web UI embedding.

Alternative	Why Not Chosen
Power BI/Fabric	Vendor lock-in, deployment overhead
Frontend-only charts	Hard reproducibility
Warehouse + jobs	Too heavy for read-only library

Structure

```
analysis/  
  datasets/                                # Source CSV files  
    heart_attack_prediction_dataset.csv  
    heart_disease_uci.csv  
    heart_disease.csv  
    heart_statlog_cleveland_hungary_final.csv  
  code_solution/                          # Analysis scripts  
    heart_attack_prediction_dataset.py  
    heart_disease_research2.py  
    heart_disease_uci.py  
    heart_statlog_cleveland_hungary_final.py
```

Data Sources

Original Datasets (Kaggle)

Dataset	Source Link
Heart Statlog Cleveland Hungary	https://www.kaggle.com/code/zohaib123/heart-disease-prediction-research-work/notebook?select=heart_statlog_cleveland_hungary_final.csv
Heart Disease UCI	https://www.kaggle.com/code/zohaib123/heart-disease-prediction-research-work/notebook?select=heart_disease_uci.csv
Heart Disease	https://www.kaggle.com/code/zohaib123/heart-disease-prediction-research-work/notebook?select=heart_disease.csv
Heart Attack Prediction	https://www.kaggle.com/code/zohaib123/heart-disease-prediction-research-work/notebook?select=heart_attack_prediction_dataset.csv

Interactive Analysis Scripts (Google Colab)

Analysis	Colab Link
Heart Statlog Cleveland Hungary	https://colab.research.google.com/drive/139hZHDxAnGv8CS3GV-HY6hkr-P7dZg8B?usp=sharing
Heart Disease Research	https://colab.research.google.com/drive/1WnrHF8mhSjNdPGSoEicCKyn
Heart Attack Prediction	https://colab.research.google.com/drive/1Cvhrqh39Pa_IFBUay20E6-SD9hs5Uuk9?usp=sharing
Heart Disease UCI	https://colab.research.google.com/drive/1_cl0Nq0X2drn6Esprw4-jaOzT1m7Qjd?usp=sharing

Analysis Methodology

Fields Analyzed

Clinical Indicators: - Age, Sex, Chest Pain Type, Resting Blood Pressure - Serum Cholesterol, Fasting Blood Sugar, Resting ECG - Maximum Heart Rate, Exercise-Induced Angina - ST Depression (Oldpeak), ST Slope - Number of Major Vessels, Thalassemia

Lifestyle Factors: - Smoking Status, Obesity, Alcohol Consumption - Exercise Hours Per Week, Physical Activity Days - Diet (Healthy/Average/Unhealthy), Sedentary Hours - Sleep Hours Per Day, Stress Level (1-10)

Medical History: - Diabetes, Family History of Heart Disease - Previous Heart Problems, Medication Use - BMI, Triglyceride Levels

Demographic Data: - Country, Continent, Hemisphere, Income Level

Statistical Methods Applied

Method	Purpose
Descriptive Statistics	Mean, median, std, min/max for all numeric features
Data Quality Checks	Missing values analysis, duplicate detection, validity checks
Target Distribution	Class balance assessment (disease vs. no disease)
Univariate Analysis	Histograms for numeric features, value counts for categorical
Bivariate Analysis	Grouped statistics by target (mean/median/std)
Welch's t-test	Compare numeric feature means between disease groups
Cohen's d	Effect size calculation for practical significance
Chi-square Test	Association between categorical features and target
Correlation Analysis	Pearson correlation matrix, correlation with target variable

Analysis Workflow

1. **Data Loading & Overview** - Load CSV, inspect shape, columns, data types
2. **Data Quality Checks** - Missing values per column, duplicate rows, value range validation
3. **Target Distribution** - Class balance, disease prevalence rate
4. **Univariate Analysis** - Distribution of each feature independently
5. **Bivariate Analysis (Numeric)** - Compare feature distributions by disease status
6. **Bivariate Analysis (Categorical)** - Disease rates within each category
7. **Statistical Testing** - t-tests for numeric, chi-square for categorical features
8. **Correlation Analysis** - Feature correlations and relationship with target
9. **Visualization** - Histograms, box plots, bar charts, correlation heatmaps

Key Findings

Statistically Significant Predictors ($p < 0.05$): - ST Depression (Oldpeak): Higher in heart disease patients - Maximum Heart Rate: Lower in heart disease patients - Age: Higher in heart disease patients - Chest Pain Type: Asymptomatic patients have highest disease rate - Exercise-Induced Angina: Strong association with heart disease - ST Slope: Flat/downsloping segments indicate higher disease prevalence

Requirements

#	Requirement	Status
1	Research code in repo	Done
2	Datasets included	Done
3	Standard Python toolchain	Done
4	Exportable artifacts	Done
5	Not coupled to runtime	Done

Limitations

Limitation	Solution
Dependencies not pinned	Add analysis/requirements.txt
No single outputs folder	Standardize analysis/outputs/

Conclusion

The research pipeline provides a comprehensive cardiovascular disease analysis framework with four distinct dataset analyses covering global health indicators and clinical predictors.

Key Achievements: - **4 datasets analyzed** covering 20,000+ patient records from multiple sources (Cleveland, Hungarian, Swiss, Statlog) - **Standardized analysis workflow:** data quality checks, target distribution, univariate/bivariate analysis, statistical testing - **Key clinical features examined:** age, cholesterol, blood pressure, heart rate, chest pain type, exercise angina, ST depression - **Statistical methods:** Chi-square tests, Mann-Whitney U tests, correlation analysis, effect size calculations - **Visualization outputs:** histograms, box plots, correlation heatmaps, bar charts for categorical distributions - **Google Colab integration:** scripts linked to interactive notebooks for reproducibility

Datasets Summary: | Dataset | Records | Features | Target Variable | |-----|-----|-----|-----|
-| | Heart Attack Prediction | 8,763 | 26 | Heart Attack Risk (binary) | | Heart Disease Research | 10,000 | 20 | Heart Disease Status (Yes/No) | | Heart Disease UCI | ~920 | 14 | Multi-class heart disease | | Statlog Combined | 1,190 | 11 | Heart disease (binary) |

This implementation supports evidence-based insights for the CVD knowledge base with transparent, reproducible analysis.

Testing & Quality Assurance

Decision

Status: Accepted | **Date:** 2026-01-05

Jest for unit tests (backend + frontend) focused on error handling/config/utilities. Playwright for E2E smoke coverage.

Alternative	Why Not Chosen
No tests	Regressions likely
Only E2E	Too slow/brittle
Only unit	Miss integration issues

Structure

backend/tests/
frontend/tests/

Requirements

#	Requirement	Status
1	Backend unit tests	Done
2	Frontend unit test setup	Done
3	E2E test harness	Done

Limitations

Limitation	Solution
Limited user story coverage	Add tests as routes complete

Conclusion

Testing Coverage

As of the latest run, the project achieves **94.02% line coverage** and **93.65% statement coverage** across all source files.

Key Coverage Highlights: - **Controllers:** 92.3% statement coverage (base controller fully covered on a line level, with partial branch and function coverage due to conditional flows) - **Disease module:** 90.22% statement coverage (controller well covered; repository and service layers partially covered with several uncovered edge and error-handling paths) - **Source module:** 97.95% statement coverage (comprehensive controller, repository, and service coverage with minimal uncovered service logic) - **Config & Constants:** 100% coverage for `types.ts`, `configService.ts`, `keys.ts`, and `messages.ts`

FAQ & Troubleshooting

FAQ

General

Q: What is this platform for?

A: A read-only CVD knowledge platform with disease entries (symptoms, risk factors, prevention) and source attribution. For education only.

Q: Is this medical advice?

A: No. Does not replace professional medical advice, diagnosis, or treatment.

Q: What data is stored about me?

A: None. The MVP avoids collecting PHI/PII. Only reference content and system telemetry are stored.

Q: Which languages are supported?

A: English (**en**) and Russian (**ru**) via the `locale` parameter.

Access & Features

Q: Do I need an account?

A: No. Read-only browsing requires no login.

Q: How do I search diseases?

A: Use the search bar or filters (symptom/risk factor). API: `GET /api/diseases?search=...`

Q: Why “No disease found” with HTTP 200?

A: Query returned zero results. This is expected behavior, not an error.

Q: What are “primary” vs “secondary” factors?

A: Primary = core symptoms; Secondary = related factors. Supports structured filtering.

Developer

Q: What are the local ports?

A: API: `localhost:4000/api` | Swagger: `localhost:4000/api-docs` | Frontend: `localhost:5173`

Q: How do I run locally?

A: Use Docker Compose (API + PostgreSQL + Redis) or run backend/frontend separately with `npm run dev`.

Q: How do I reset the database?

A: Run migrations + seed scripts. See [DB Schema](#).

Troubleshooting

Problem	Cause	Solution
Blank screen	Frontend not running	Start frontend, check devtools console
<code>ERR_CONNECTION_REFUSED</code>	Backend not running	Start backend, verify <code>localhost:4000/api/health</code>
CORS error	Origin mismatch	Match <code>CORS_ORIGIN</code> to frontend URL
“No disease found”	Empty DB or strict filters	Run seed; remove filters
Swagger not loading	Backend down	Check <code>localhost:4000/api/health/details</code>
DB connection error	Wrong <code>DATABASE_URL</code>	Verify Postgres container and credentials
Redis error	Redis not running	Check Redis container; use <code>redis</code> host in Docker

Error Messages

Message	Meaning	Fix
<code>No disease found (200)</code>	Zero results	Relax filters, verify seed

Message	Meaning	Fix
Bad Request (400)	Invalid parameters	Check types (<code>skip >= 0, 1 <= take <= 100</code>)
Resource not found (404)	Invalid route	Verify path matches API Reference
Internal Server Error (500)	Backend error	Check logs

Getting Help

- [Documentation](#) | [API Reference](#)
- Swagger: <http://localhost:4000/api-docs>
- Issues: <https://github.com/Doshirac/cvd-platform/issues>

Bug reports should include: steps to reproduce, expected vs actual behavior, screenshots, browser/OS info.

Feature Walkthrough

Core user flows of the CVD Platform.

[Original UI/UX Documentation](#)

Feature 1: Navigation, Theme & Language

Use the header to navigate sections and adjust preferences.

How to Use: 1. Click **Home**, **Sources**, or **Research** (mobile: burger menu) 2. Toggle theme via the icon — persists across sessions 3. Select language via globe icon

[Header navigation](#) | [Mobile nav](#) | [Desktop nav](#)

Tip: If theme seems stuck, hard-refresh and check localStorage.

Feature 2: Disease Library (Home)

Browse cardiovascular diseases as standardized cards.

How to Use: 1. Open **Home** from header 2. Scroll/paginate through disease list 3. Click a card to view details (if enabled)

[Disease library](#) | [Navigation flow](#)

Feature 3: Search & Filter

Find diseases by symptoms, risk factors, or keywords.

How to Use: 1. Enter keywords in the search bar 2. Apply **Symptom** and/or **Risk Factor** filters 3. Use **Alphabet filter** (A–Z) if available 4. Click **See more** to load additional results

[Search/filters](#) | [Filter panel](#) | [Mobile filters](#)

API: GET /api/diseases?search=...&symptom=...&riskFactor=...&skip=0&take=10&locale=en

Feature 4: Sources & Research

View attribution sources and research summaries.

How to Use: 1. Navigate to **Sources** — click **View Resource** to open external links 2. Navigate to **Research** — browse research cards (if enabled)

[View source](#) | [Research cards](#)

Additional Features (if enabled)

Feature	Description
Back to Diseases	Quick return button — Image
Scroll-to-Top	Return to top on long pages — Image
Tooltip Badges	Hover/long-press codes for full names — Image

Keyboard Shortcuts

Shortcut	Action
Ctrl/Cmd + R	Refresh page
Ctrl/Cmd + F	Find on page
Alt + ←	Navigate back
Esc	Close dropdown/menu

Feature Comparison

Feature	MVP	Post-MVP
Browse/search/filter diseases		
Disease details page	In progress	
Sources & attribution		
Research Cards	May be disabled	

User Guide

Practical guide to using the CVD Platform for browsing and local development.

Contents

- [Features Walkthrough](#)
 - [FAQ & Troubleshooting](#)
-

System Requirements

Requirement	Minimum	Recommended
Browser	Chrome/Edge/Firefox/Safari (latest 2 versions)	Latest
Resolution	1280×720	1920×1080
Internet	Required	Stable
Device	Desktop or mobile	Desktop

Accessing the Application

1. Open your web browser
2. Navigate to the application URL:
 - **Production:** (if deployed) use your deployed URL
 - **Local development:** start the frontend and open `http://localhost:5173`
3. If you are running locally, ensure the API is available at `http://localhost:4000/api`

Run locally (recommended)

1. Start the stack with Docker Compose (API + DB + Redis).
2. Seed the database.
3. Start the frontend dev server.

Local URLs: - Frontend: `http://localhost:5173` - API: `http://localhost:4000/api` - Swagger: `http://localhost:4000/api-docs`

First Launch Checklist

1. Verify header/navigation renders
2. Toggle theme — confirm persistence
3. Browse disease list, verify loading states

Quick Start

Task	Action
Browse diseases	Open Home , scroll list
Search/filter	Use search bar and filters
Verify sources	Open Sources , click external links
Check health	GET <code>/api/health/details</code>

API Verification

Use Swagger (`/api-docs`) or call directly:

```
GET /api/health/details
GET /api/diseases?take=10&skip=0&locale=en
GET /api/diseases?search=heart&locale=en
GET /api/sources?take=10&skip=0
```

User Roles

Role	Access
Visitor	Read-only browsing, theme/language preferences
Developer	Local services, DB seeding, Swagger access

API Reference

Base URL: `http://localhost:4000/api`

Endpoints Summary

Method	Endpoint	Description
GET	<code>/health</code>	Health check (204)
GET	<code>/health/details</code>	Detailed health info
GET	<code>/diseases</code>	List diseases (paginated, filterable)
GET	<code>/diseases/risk-factors</code>	List all risk factors
GET	<code>/diseases/by-letter</code>	Filter diseases by initial letter
GET	<code>/sources</code>	List data sources (paginated)

Common Parameters

- `skip` / `take` — pagination (default: 0 / 6)
- `locale` — "en" or "ru" (default: "en")
- `search` — free-text search

Response Format

Success: Array of objects or `{ "message": "No [resource] found." }`

Error:

```
{ "success": false, "message": "...", "errors": [] }
```

Code	Meaning
200	OK
204	No Content
400	Bad Request
404	Not Found
500	Server Error

Full Documentation

- OpenAPI/Swagger: [GitBook](#)
- OpenAPI JSON: `backend/openapi.json`

Database Schema

Attribute	Value
Database	PostgreSQL
ORM	Prisma
Schema	backend/prisma/schema.prisma

ER Diagram

[View Diagram](#)

Core Tables

Table	Purpose
Disease	Primary disease entities (code, name, description, prevention)
Symptom	Standardized symptoms/signs
RiskFactor	Risk and protective factors
DiseaseSymptom	Disease-Symptom join (priority, typicality)
DiseaseRiskFactor	Disease-RiskFactor join (direction)
Source	Data sources (name, link)
*_translation	Localized content (EN/RU)

Enums

- **Locale:** en, ru
- **SymptomCategory:** sign, symptom
- **SymptomPriority:** primary, secondary
- **Typicality:** typical, possible, rare
- **RiskDirection:** risk, protective

Setup

```
cd backend
npm run migrate
npm run generate
npm run seed
```

Full Documentation

- **Detailed Schema:** [Google Doc](#)
- **Prisma Schema:** backend/prisma/schema.prisma

Glossary

Key Terms

Term	Definition
CVD	Cardiovascular Disease - core domain of the platform
Disease card	Structured disease entry (name, symptoms, prevention)
Symptom	Patient-reported/observed manifestation
Risk factor	Factor increasing disease probability

Term	Definition
Protective factor	Factor reducing disease probability
Locale	Language support (EN/RU)
MVP	Minimum Viable Product

Key Acronyms

Acronym	Meaning
API	Application Programming Interface
ICD-10	International Classification of Diseases
SNOMED	Clinical terminology standard
ORM	Object-Relational Mapping (Prisma)
i18n	Internationalization
GDPR	General Data Protection Regulation

Tech Stack Terms

Term	Description
Express	Node.js REST API framework
Vite	Frontend build tool
Prisma	Database ORM
Redis	Caching layer
Playwright	E2E testing
Storybook	UI component docs

Retrospective

This section reflects on the development of the **CVD Platform**, summarizing results, limitations, and lessons learned.

Project Status Overview

The project reached a **functional MVP state** with:

- **Working backend** with layered architecture (controllers → services → repositories)
- **Normalized data model** with bilingual support (EN/RU) and junction tables for symptoms/risk factors
- **Docker Compose** setup for reproducible local development (API + PostgreSQL + Redis)
- **Testing infrastructure** on both ends (Jest for backend, Playwright for e2e)
- **Frontend partially scaffolded** — some routes incomplete due to late requirements

What Went Well

Technical Successes

- Layered backend architecture with Prisma + PostgreSQL provided predictable API behavior
- Docker Compose enabled consistent cross-machine development
- Normalized schema (translations, junction tables) supported bilingual “disease cards”
- Early observability setup (Winston/Morgan, Sentry) reduced debugging time
- Testing foundation exists for future reliability improvements

Process Successes

- Read-only MVP scope reduced compliance risk (no user accounts, no PHI/PII)
- Documentation treated as deliverable, aligned to actual codebase
- Incremental development: backend/data model stabilized before UI work

Personal Achievements

- Built full-stack system (React + Vite, Express + Prisma, PostgreSQL + Redis, Docker)
- Improved data modeling skills (migrations, seeding, normalized entities)
- Strengthened ability to translate academic requirements into implementable MVP

What Didn't Go as Planned

Planned	Actual Outcome	Cause	Impact
Fully working frontend pages	Partially scaffolded; some routes show Not Found	Late frontend requirements from teachers	Critical
Complete Disease Card UI	Backend ready, UI not finished end-to-end	Time constraints; prioritized backend + docs	High
Full EN/RU localization toggle	Backend supports locale; UI switch incomplete	Limited time for i18n flow	Medium

Key Challenges

1. **Late Frontend Requirements** — Requirements clarified after deadline; focused on stabilizing back-end and documentation instead
2. **Documentation Drift** — Templates easily diverged from implementation; resolved by re-checking controllers/OpenAPI

Technical Debt & Known Limitations

ID	Issue	Severity	Potential Fix
TD-001	Frontend routes not fully wired	High	Complete route config; add e2e coverage
TD-002	Inconsistent DI patterns in backend	Medium	Standardize to one approach
TD-003	API returns message instead of empty array	Low	Return empty arrays + metadata

Future Development Directions

High Priority

1. **Complete frontend flows** — Wire main routes to real endpoints with proper loading/error states
2. **Robust localization** — Ensure language switch controls `locale` in all API requests

Medium Priority

3. **Standardize API contracts** — Consistent response shapes for list endpoints
4. **Expand test coverage** — Playwright for happy paths; backend integration tests

Nice to Have

5. SEO metadata and sitemap
6. Cache TTL tuning and invalidation strategy
7. Formalize Python analysis pipeline

Lessons Learned

Technical

Lesson	Application
Normalized schema pays off even for MVPs	Start with clean relational model; avoid over-denormalization
Standardized error/empty-state contracts matter	Treat response shapes as product contract
Reproducible environments reduce risk	Containerize early; document env vars

Process

Lesson	Application
Requirements need a freeze point	Agree on MVP requirement freeze date
Documentation alongside implementation	Keep docs close to code changes

What Would Be Done Differently

Area	Change	Why
Planning	Set “end-to-end MVP demo” milestones earlier	Prevents UI lagging behind backend
Technology	Enforce single DI approach	Reduces maintenance overhead
Process	Formal requirements sign-off checkpoint	Reduces late changes and scope creep
Scope	Cut to “core reading flows” first	Higher probability of complete UX

Skills Developed

Skill	Before	After
Full-stack delivery (FE/BE/DB)	Intermediate	Advanced
Database modeling (Prisma/PostgreSQL)	Beginner–Intermediate	Advanced
Testing (Jest/Playwright/Storybook)	Beginner	Intermediate

Key Takeaways

1. A coherent MVP requires an end-to-end “happy path” early

2. Data modeling decisions strongly determine maintainability
3. Clear contracts reduce confusion and ease thesis evaluation

Retrospective completed: 2026-01-05

Final Conclusion

This thesis explored the development of a web-based knowledge system for cardiovascular disease prevention, designed to consolidate fragmented CVD information into standardized, referenced disease cards with consistent descriptions, risk factors, symptoms, and practical prevention guidance.

Achieved Results

During the course of the project, the following results were achieved:

1. Backend Architecture & API Delivery

A production-ready, modular API for cardiovascular disease data was delivered with focus on reliability, observability, and developer experience. The implementation features a modular monolith architecture with clear domain boundaries (disease, source, cache, monitoring) and dependency injection. Full containerization via Docker with multi-stage builds, health checks, resource limits, and graceful shutdown was achieved. Comprehensive observability includes structured JSON logging, Sentry error tracking, health endpoints, and centralized error handling. High test coverage (87%+ lines) with Jest, enforced code quality via ESLint/Prettier/Husky pre-commit hooks ensures maintainability. Bilingual support through Prisma translation tables, Redis caching for performance, and OpenAPI/Swagger documentation complete the backend implementation.

2. Database Design & Data Integrity

The Cardiovascular Disease Knowledge Base database provides a robust, normalized foundation for clinical and public-health data. Key achievements include a comprehensive entity structure (diseases, symptoms, risk factors), strict referential integrity with clear indexes and constraints, bilingual (localized) design via translation tables and SQL views, automated versioning and migrations through Prisma and Git, and extensibility for future data layers (raw → staging → mart). This implementation supports consistent, query-optimized, and language-aware access to cardiovascular disease data.

3. Research Pipeline & Analytics Artifacts

A comprehensive cardiovascular disease analysis framework was established with four distinct dataset analyses covering global health indicators and clinical predictors. The pipeline analyzed 4 datasets covering 20,000+ patient records from multiple sources (Cleveland, Hungarian, Swiss, Statlog). A standardized analysis workflow includes data quality checks, target distribution, univariate/bivariate analysis, and statistical testing. Key clinical features examined include age, cholesterol, blood pressure, heart rate, chest pain type, exercise angina, and ST depression. Statistical methods applied include Chi-square tests, Mann-Whitney U tests, correlation analysis, and effect size calculations.

4. Deployment & DevOps

A production-ready Docker Compose architecture was implemented with three core services: backend API (cvd-api), PostgreSQL database (db-1), and Redis cache (redis-1). Multi-stage Dockerfile reduces API image size from ~1.5 GB to 891 MB. Named volumes ensure data persistence across container restarts. Health checks monitor service availability every 30 seconds with automatic recovery. Resource limits prevent resource exhaustion on shared infrastructure. Non-root user execution enhances security posture.

5. Testing & Quality Assurance

The project achieves **94.02% line coverage** and **93.65% statement coverage** across all source files. Controllers achieve 92.3% statement coverage, Disease module reaches 90.22% statement coverage, Source module achieves 97.95% statement coverage, and Config & Constants achieve 100% coverage.

6. Observability Implementation

Comprehensive monitoring and debugging capabilities were implemented for both frontend and backend systems. Structured logging uses Winston with configurable log levels and JSON-formatted output. Error tracking via Sentry provides real-time exception monitoring with automatic capture and performance monitoring. Global error handling through centralized Express middleware provides structured JSON responses with custom ApiError class for semantic HTTP status codes.

7. Frontend Foundation

A modern, maintainable React SPA was delivered for browsing cardiovascular disease content. The implementation uses React 19 + TypeScript + Vite stack with fast HMR and optimized builds. Redux Toolkit provides centralized state management. 13+ reusable UI components were created. Dark/Light theming, Error Boundary integration, Playwright E2E tests, Storybook documentation, and code quality tooling complete the frontend foundation.

Limitations and Non-Implemented Components

Despite the achieved results, several components were intentionally not implemented within the scope of the diploma project:

1. Frontend Route Completion

Core frontend pages (MainPage, SourcePage, DiseasePage) are not fully wired to backend endpoints. This limitation is due to late frontend requirements clarification from teachers, which arrived after the deadline. The focus was shifted to stabilizing backend and documentation instead.

2. Complete Internationalization Toggle

While the backend supports locale-aware responses (EN/RU) and translation tables exist in the database, the UI language switch is incomplete. Limited time prevented full i18n flow implementation across all user-facing components.

3. Source-Disease Relationship

Sources are not directly linked to individual diseases in the current data model. Future enhancement would add a DiseaseSource relation for per-disease attribution and citation tracking.

4. Advanced Search Features

Full-text search may experience performance limitations with larger datasets. Future implementation would add tsvector + GIN indexes for optimized PostgreSQL full-text search capabilities.

Temporal Scope and Validity

As a result of these constraints, the developed solution represents a stable and reproducible MVP of the CVD knowledge platform, demonstrating the core architectural patterns and data structures required for a comprehensive cardiovascular disease prevention system. The platform is designed for future extensibility and can evolve with additional features as requirements mature.

Overall Assessment

The Cardiovascular Disease Prevention Platform successfully demonstrates the viability of a web-based knowledge system for consolidating fragmented CVD information. The project delivers:

- **A complete backend API** with layered architecture, Redis caching, and bilingual support
- **A normalized database schema** supporting diseases, symptoms, risk factors, and translations
- **A research pipeline** with 4 analyzed datasets covering 20,000+ patient records
- **Production-ready containerization** with Docker Compose, health checks, and resource management
- **High code quality** with 94%+ test coverage and enforced linting standards
- **Comprehensive observability** through structured logging and Sentry error tracking

The platform provides query-optimized, language-aware access to cardiovascular disease data and establishes a foundation for integration with frontend applications, analytics dashboards, and future microservices expansion. While some frontend features remain incomplete due to timeline constraints, the core knowledge system architecture and data foundation are fully operational and ready for continued development.

Final Conclusion completed: 2026-01-07